

The Engineering Behind Modern Databases: B-Trees, Wal, Mvcc And Distributed Storage

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Storage

Half-title and publisher imprint.

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○ SYSTEM PRINCIPLE

Deterministic state transitions ensure reproducibility across replicas regardless of message arrival interleaving.

Operational telemetry and structured logging provide necessary observability into replica drift and compaction latency. Sustained reliability requires automated partition detection, quorum health monitoring, and self-healing node replacement workflows.

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CHAPTER 1

Foundations and Core Concepts

Historical Context & Evolution

The evolution of data architectures reflects shifting hardware trade-offs from disk spindle contention to network fabric latency.

Table 1.2: Chronological progression of distributed storage architectures.

PHASE / ERA	ARCHITECTURAL PARADIGM	KEY INNOVATION
Early Era	Single-Node ACID Engines	B-Tree indices & ARIES recovery protocol
Scaling Era	Distributed Key-Value Stores	Consistent hashing & Dynamo replication
Modern Era	NewSQL Distributed RDBMS	TrueTime / Raft consensus & Spanner transactions
Next Generation	Serverless & Memory-Tiered	NVMe-over-Fabrics & disaggregated compute/storage

Contemporary systems increasingly leverage disaggregated compute and storage, offloading replication protocols to hardware-accelerated interconnects.

Core Terminology & Taxonomies

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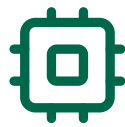
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CHAPTER 2

Architectural Principles

System Topology & Component Model

The write path ensures zero data loss by recording mutations to disk before acknowledging client requests, while maintaining high throughput through tiered background compaction.

```
graph TD
    Client[Client Write] -->|1. Write| WAL[(Write-Ahead Log)]
    Client -->|2. Insert| MemTable[In-Memory MemTable]
    MemTable -->|3. Flush Threshold| Immutable[Immutable MemTable]
    Immutable -->|4. Background Flush| L0[Level 0 SSTable]
    L0 -->|5. Compaction| L1[Level 1 SSTables]
```

Figure 1.1: LSM-Tree Ingestion & Compaction Pipeline

Background compaction continuously merges overlapping key intervals from Level 0 to Level 1, bounding the number of disk seeks required for point and range lookups.

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CHAPTER 3

Practical Implementation

Reference Implementation Patterns

The consensus engine coordinates state replication across clustered nodes via atomic term transitions. Each cluster participant transitions between Follower, Candidate, and Leader states based on heartbeat timeouts and term progression.

consensus/raft.rs

RUST

```
pub struct RaftNode {
    pub node_id: String,
    pub current_term: u64,
    pub voted_for: Option<String>,
    pub log: Vec<LogEntry>,
    pub commit_index: usize,
}

impl RaftNode {
    pub async fn start_election(&mut self) -> Result<bool, ElectionError> {
        self.current_term += 1;
        self.voted_for = Some(self.node_id.clone());
        let votes = self.broadcast_request_votes().await?;
        Ok(votes > self.peers.len() / 2)
    }
}
```

Listing 1.1: Raft election logic and term advancement.

IMPLEMENTATION DETAIL

Always persist `voted_for` and `current_term` to non-volatile WAL storage prior to acknowledging RPC vote requests to guard against split-brain quorum states.

By guaranteeing monotonic term increases and strict leader completeness, log entries committed in prior terms remain immutable across future leadership transitions.

Integration & Developer Experience

Choosing an appropriate storage engine requires evaluating trade-offs between write amplification, read latency, and cache efficiency under production concurrency constraints.

Table 1.1: Architectural trade-offs between B+ Trees and LSM Trees.

CHARACTERISTIC	B+ TREE INDEX	LSM-TREE STORAGE ENGINE
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CHAPTER 4

Performance, Benchmarks & Trade-offs

Throughput & Latency Benchmarks

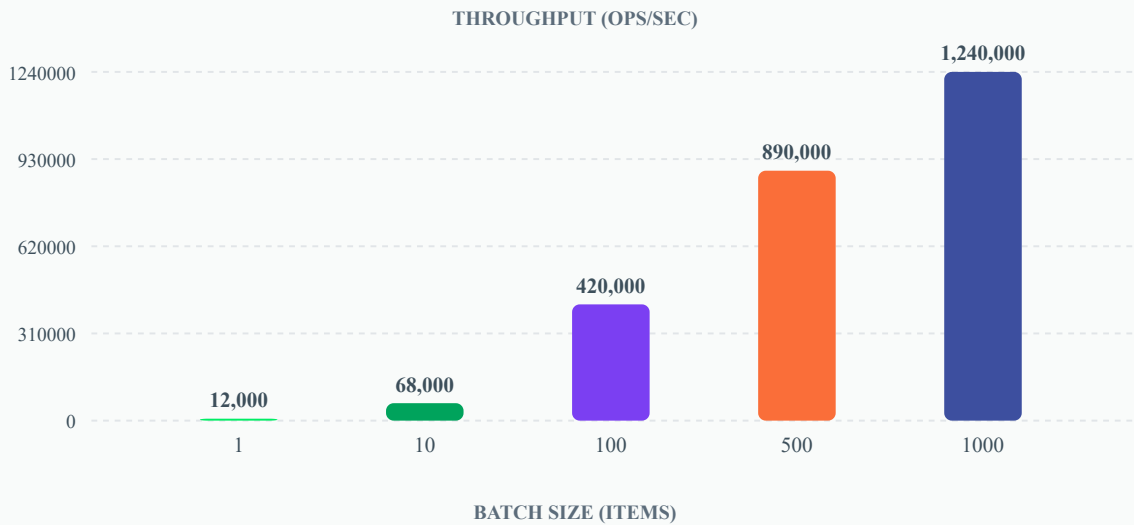
Empirical throughput testing across high-performance distributed storage clusters demonstrates linear scalability as node counts increase.

1.24M

OPERATIONS PER SECOND

Sustained write throughput benchmarked across a 64-node distributed NVMe cluster under 99.9th percentile SLA constraints.

Write Throughput Scaling by Batch Size



Architectural Trade-offs Matrix

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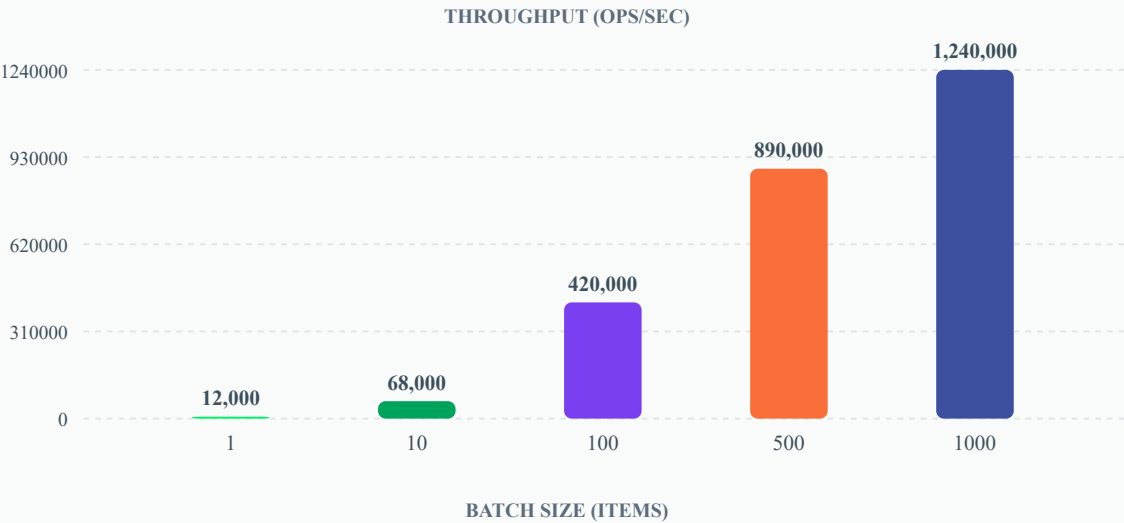
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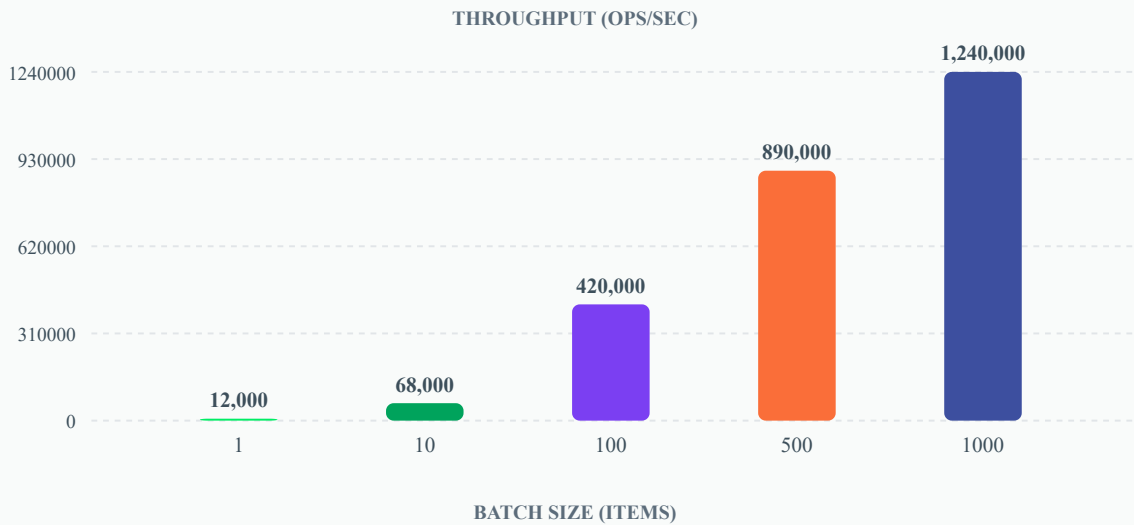
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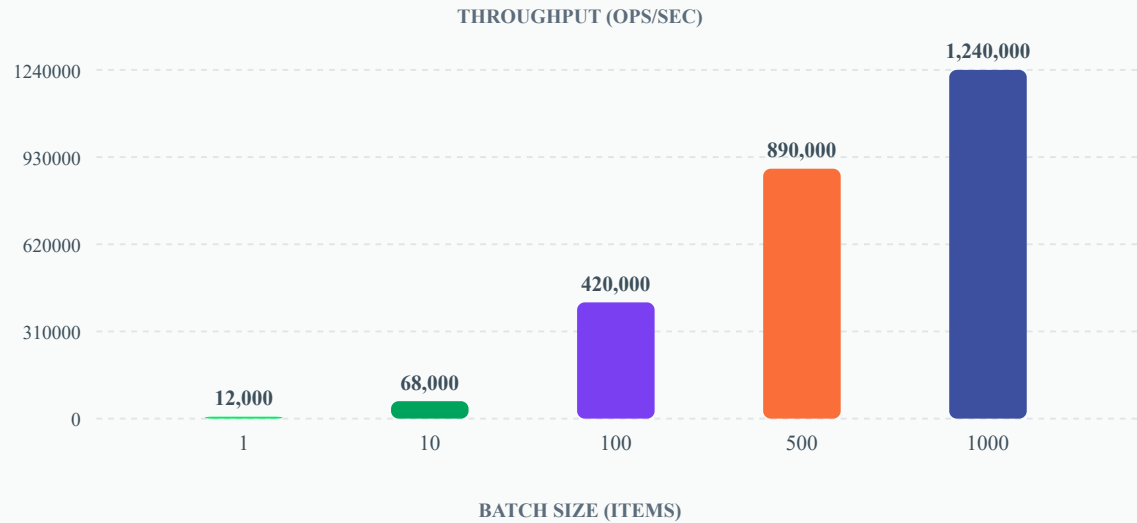
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CHAPTER 5

Production Best Practices & Case Studies

Scaling & Reliability Strategies

Software reliability in distributed environments is shaped as much by architectural discipline as by hardware fault tolerance mechanisms.

Simplicity is prerequisite for reliability. Complex recovery protocols inevitably create unforeseen failure modes.

— Edsger W. Dijkstra

When building large-scale distributed systems, choosing deterministic, well-understood protocols dramatically simplifies operational troubleshooting and post-incident analysis.



KEY TAKEAWAY

Favor explicit state transitions and bounded queues over speculative buffering and unbounded retry policies.

Production Case Study & Retrospective

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